

Unit-4

- **PHP and MySQL Basic commands** with PHP examples -
Connection to server - creating database - selecting a
database-listing database-listing table names-creating a table,-
inserting data-altering tables – queries -deleting database
-deleting data and tables -PHP myadmin and database bugs.

- <?php
- \$servername = "localhost";
- \$username = "root";
- \$password = "";
- \$dbname = "t_db";
-
- // Create connection
- \$conn = new mysqli(\$servername, \$username, \$password, \$dbname);
-
- // Check connection
- if (\$conn->connect_error) {
- die("Connection failed: " . \$conn->connect_error);
- }
- echo "Connected successfully";
- ?>

- `<?php` — Starts the PHP code.
- `$servername = "localhost";` — The server name where MySQL is running.
- "localhost" means the database server is on the same computer as your PHP code.
- `$username = "root";` — The MySQL username used to connect.
- "root" is the default MySQL admin user.
- `$password = "";` — The password for the MySQL user.
- Empty here (no password set for root).
- `$dbname = "t_db";` — The name of the database you want to connect to.

- `$conn = new mysqli($servername, $username, $password, $dbname);`
- Explanation:
 - This line creates a new connection object (\$conn) using the mysqli class.
 - It passes the four parameters:
 - Server name
 - Username
 - Password
 - Database name
 - \$conn will now represent your connection to the database.

- `// Check connection`
- `if ($conn->connect_error) {`
- `die("Connection failed: " . $conn->connect_error);`
- `}`
- Explanation:
- This checks whether the connection was successful.
- `$conn->connect_error` — if there is an error, it stores the error message.
- `if ($conn->connect_error)` — means "if there's any connection error".
- `die("Connection failed: " . $conn->connect_error);` — stops the script and prints the error message.

- `echo "Connected successfully";`
- `?>`
- Explanation:
- If no error occurs, this message is printed — confirming the connection was successful.
- `?>` — ends the PHP code.

- Summary:
- This script connects your PHP program to the MySQL database named t_db using:
- Server: localhost
- User: root
- Password: (blank)
- Database: t_db

CREATE

```
• <?php
• $servername = "localhost";
• $username = "root";
• $password = "";
• $dbname = "t_db";
•
• // Create connection
• $conn = new mysqli($servername, $username, $password, $dbname);
•
• $name = 'John Doe';
• $address = '123 Street';
• $salary = 50000;
•
• $sql = "INSERT INTO employees (name, address, salary) VALUES ('$name', '$address', $salary)";
• if (mysqli_query($conn, $sql)) {
•     echo "Record added successfully";
• } else {
•     echo "Error: " . mysqli_error($conn);
• }
• ?>
•
```


Delete

```
• <?php
• $servername = "localhost";
• $username = "root";
• $password = "";
• $dbname = "t_db";
•
• // Create connection
• $conn = new mysqli($servername, $username, $password, $dbname);
•
• // Check connection
• if ($conn->connect_error) {
•     die("Connection failed: " . $conn->connect_error);
• }
•
• // Specify the name to delete
• $name = 'Deepa';
•
• // SQL query to delete record by name
• $sql = "DELETE FROM employees WHERE name = '$name'";
•
```

- // Execute the query
- if (mysqli_query(\$conn, \$sql)) {
- echo "Record with name '\$name' deleted successfully";
- } else {
- echo "Error deleting record: " . mysqli_error(\$conn);
- }

- // Close connection
- \$conn->close();
- ?>

Update

```
• <?php
• $servername = "localhost";
• $username = "root";
• $password = "";
• $dbname = "t_db";
•
• // Create connection
• $conn = new mysqli($servername, $username, $password, $dbname);
•
• // Check connection
• if ($conn->connect_error) {
•     die("Connection failed: " . $conn->connect_error);
• }
•
• // Values to update
• $name = 'Deepa';    // Employee name to identify the record
• $new_address = '456 Avenue';
• $new_salary = 60000;
•
```

- // SQL query to update record
- \$sql = "UPDATE employees SET address = '\$new_address', salary = \$new_salary WHERE name = '\$name'";
- // Execute query
- if (mysqli_query(\$conn, \$sql)) {
- echo "Record updated successfully for \$name";
- } else
- {
- echo "Error updating record: " . mysqli_error(\$conn);
- }
- // Close connection
- \$conn->close();
- ?>

PHP – MySQL Connection Example

- <?php
- // Step 1: Create connection variables
- \$servername = "localhost"; // Server name (usually localhost)
- \$username = "root"; // MySQL username (default is root)
- \$password = ""; // MySQL password (empty by default in XAMPP)
- \$database = "student_db"; // Your database name
-
- // Step 2: Create connection using mysqli_connect()
- \$conn = mysqli_connect(\$servername, \$username, \$password, \$database);
-
- // Step 3: Check the connection
- if (!\$conn) {
- die("Connection failed: " . mysqli_connect_error());
- }
- echo "Connected successfully to the database!";
- ?>

Step

1. Define Connection Variables

2. Create Connection

3. Check Connection

4. Close Connection (Optional)

Description

You specify the MySQL **server name**, **username**, **password**, and **database name**.

`mysqli_connect()` function is used to connect PHP to the MySQL server.

The `if` condition checks if the connection fails. If yes, it shows an error using `mysqli_connect_error()`. Otherwise, it displays a success message.

You can close the connection using `mysqli_close($conn)`; after performing queries.

- **Example Output**

- If the connection is successful:
- Connected successfully to the database!
- If it fails:
- Connection failed: Access denied for user 'root'@'localhost'

- You must create the database (e.g., student_db) in phpMyAdmin before connecting.
- If using WAMP, start Apache and MySQL before running the code.
- This connection can be used to insert, update, delete, or fetch data once established.

CRUD (Create, Read, Update, Delete) example using PHP and MySQL

- **1. Create Database and Table**
- Before writing PHP, open **phpMyAdmin** and run this SQL:
- CREATE DATABASE student_db;
- USE student_db;
- CREATE TABLE students (
 - id INT AUTO_INCREMENT PRIMARY KEY,
 - name VARCHAR(100),
 - email VARCHAR(100),
 - department VARCHAR(50)
-);

- PHP CRUD Application Code
- Save the following as crud.php inside your www folder (for WAMP).

- <?php
- // Step 1: Connect to Database
- \$conn = mysqli_connect("localhost", "root", "", "student_db");
- if (!\$conn) {
- die("Connection failed: " . mysqli_connect_error());
- }
-
- // Step 2: Handle Create Operation
- if (isset(\$_POST['add'])) {
- \$name = \$_POST['name'];
- \$email = \$_POST['email'];
- \$dept = \$_POST['dept'];
- \$sql = "INSERT INTO students (name, email, department) VALUES ('\$name', '\$email', '\$dept')";
- mysqli_query(\$conn, \$sql);
- }

- // Step 3: Handle Delete Operation
- if (isset(\$_GET['delete'])) {
- \$id = \$_GET['delete'];
- mysqli_query(\$conn, "DELETE FROM students WHERE id=\$id");
- header('Location: crud.php');
- }

- // Step 4: Handle Update Operation
- if (isset(\$_POST['update'])) {
 - \$id = \$_POST['id'];
 - \$name = \$_POST['name'];
 - \$email = \$_POST['email'];
 - \$dept = \$_POST['dept'];
 - \$sql = "UPDATE students SET name='\$name', email='\$email', department='\$dept' WHERE id=\$id";
 - mysqli_query(\$conn, \$sql);
 - header('Location: crud.php');
- }
- ?>

- <!DOCTYPE html>
- <html>
- <head>
- <title>PHP MySQL CRUD Example</title>
- </head>
- <body>
- <h2>Student Management System</h2>
- <!-- Add or Update Student Form -->
- <form method="POST" action="">
- <?php
- if (isset(\$_GET['edit'])) {
- \$id = \$_GET['edit'];
- \$result = mysqli_query(\$conn, "SELECT * FROM students WHERE id=\$id");
- \$row = mysqli_fetch_assoc(\$result);
- ?>

- <input type="hidden" name="id" value="<?php echo \$row['id']; ?>">
- Name: <input type="text" name="name" value="<?php echo \$row['name']; ?>">

- Email: <input type="text" name="email" value="<?php echo \$row['email']; ?>">

- Department: <input type="text" name="dept" value="<?php echo \$row['department']; ?>">

- <button type="submit" name="update">Update</button>
- <?php } else { ?>
- Name: <input type="text" name="name" required>

- Email: <input type="text" name="email" required>

- Department: <input type="text" name="dept" required>

- <button type="submit" name="add">Add Student</button>
- <?php } ?>
- </form>

- <hr>
- <!-- Display Records -->
- <h3>Student Records</h3>
- <table border="1" cellpadding="10">
- <tr>
- <th>ID</th>
- <th>Name</th>
- <th>Email</th>
- <th>Department</th>
- <th>Action</th>
- </tr>


```
• <?php
•     $result = mysqli_query($conn, "SELECT * FROM students");
•     while ($row = mysqli_fetch_assoc($result)) {
•         echo "<tr>
•             <td>{$row['id']}</td>
•             <td>{$row['name']}</td>
•             <td>{$row['email']}</td>
•             <td>{$row['department']}</td>
•             <td>
•                 <a href='crud.php?edit={$row['id']}'>Edit</a> |
•                 <a href='crud.php?delete={$row['id']}'>Delete</a>
•             </td>
•         </tr>";
•     }
•     ?>
• </table>
• </body>
• </html>
•
```

Step	Action	Description
1	Database Connection	Connect to MySQL using <code>mysqli_connect()</code>
2	Create (Add Record)	When user submits the form with Add , an SQL INSERT query adds a new student record
3	Read (Display Records)	<code>SELECT * FROM students</code> fetches and displays all student data in a table
4	Update	When user clicks Edit , the form is filled with existing data and saved using an SQL UPDATE query
5	Delete	When user clicks Delete , an SQL DELETE query removes the student record